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An AI-driven Constraint-Based Travel Optimization System using Multi-Source Data Synthesis



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An AI-driven Constraint-Based Travel Optimization

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Writing Submission

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System using Multi-Source Data Synthesis

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I. INTRODUCTION

Abstract— The increased development of digital traveling platforms has caused overload of fragmented information and planning multi-city itineraries is more complex and inefficient. The article focuses on

SwadeshiYatra, an intelligent travel ecosystem based on AI, created to streamline the trip planning process with intelligent automation and personalization. The architecture of the system has a modern full-stack code based on Next.js, React 19, TypeScript, and Tailwind CSS that will guarantee a responsive and scalable user interface. SwadeshiYatra is fundamentally a combination of Google Gemini Large Language Models (LLMs) to allow generating itineraries based on the context, analyzing the feasibility of the issue constraint-based, and understanding it using natural language.

The suggested platform presents a new three-step pipeline which includes discovery, persona-based preference calibration and AI-driven synthesis to create time-sensitive and dynamic travel plans. In contrast to the conventional systems whose main foundation is the use of static data, SwadeshiYatra uses real-time information of Google Places and Wikipedia to make context-sensitive recommendations. One of the contributions is AI Feasibility Engine which analyses constraints like budget, travelling time, and geography distance to make itineraries realistic and feasible. Moreover, a multi-role authentication system helps improve the reliability of the system by authenticating service providers such as hotels, restaurants and guides via an administrative process. The experimental outcomes show that the system can significantly decrease the complexity of the planning process and enhance its relevance and feasibility of the itinerary. SwadeshiYatra will fill the most important gaps in the current travel technologies by helping to overcome such issues as information overload, absence of personalization, and ineffective service discovery. The system uses the progress of reasoning based on LLM, real-time information integration, constraint-driven optimization to improve decision-making and provide intelligent and user-friendly travel planning solutions. [3][4][5].

Keywords— Artificial Intelligence (AI), Large Language Models (LLMs), Multi-City Travel Planning, Constraint-Based Optimization, Semantic Route Optimization, Itinerary Generation, Feasibility Analysis, Personalized Recommendation Systems, Smart Tourism Systems, Data Integration, Google Gemini API, Real-Time Data Processing, Human-Centered AI, Role-Based Access Control (RBAC), Travel Technology.

The blistering development of online tourism platforms has brought a lot of changes in the travel planning but has also come with its own challenges namely information overload and service fragmentation. Travelers tend to use various applications to find destinations, book accommodation, and plan their routes, which leads to inefficiency and inadequate coordination of itineraries. The current systems often do not consider real life constraints like travel time, budget constraints, and schedules of attractions, resulting in impractical travel plans. Also, the absence of validated verification processes of local service providers is a matter of concern in terms of trust and reliability. These constraints underscore the fact that there is a need to have a combined, smart solution that can produce individualized and attainable travel plans in a single ecosystem. To overcome such difficulties, the present paper suggests SwadeshiYatra, an AI-based platform of individual multicity travel planning. It is developed based on the current web technologies such as Next.js, React 19, TypeScript and Tailwind CSS and uses the Gemini API of Google to provide contextual insight, generation of itineraries and feasibility analysis. The platform uses a three-step architecture that includes discovery, personas-based preference calibration and AI-based synthesis to create dynamic user-centered itineraries based on their individual travel profile. One of the main contributions is the AI Feasibility Engine that helps analyze the constraints in terms of time, cost, and distance to make travel plans realistic and achievable. Moreover, a multi-role verification system increases trust by checking on the service providers using an administrative workflow. SwadeshiYatra uses both structured data combination and sophisticated AI methods to eliminate fragmentation and lack of smart coordination in current tourism systems, providing a single, personalized, and constraint-sensitive travel planning platform..[11][15].

A. Background

Travel technology has shifted towards the old fashioned guide books into all digital with real time booking and recommendation. Current travel apps rely on web technology and data systems to assist users with their destinations and traveling. Most of the current systems are

however silo-based in the sense that they focus on a specific functionality, i.e. hotel reservations, navigation or attraction listings. The latest advancement in Artificial Intelligence, namely Large Language Models (LLMs), has brought the possibility of comprehending the context and

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creating content automatically, which opens a fresh set of possibilities to intelligent travel assistants. Despite these processes, AI is not highly developed in combination with real-time data streams and constraint-based planning. Current systems do not usually include personalization of users, feasibility testing and service provider testing into one architecture and a big gap can be seen in offering trusted and holistic travel solutions. The recent advancements in Large Language Models and smart tourism architectures have made context-aware recommendations possible, however, there are still issues with incorporating real-time information and feasibility limits. [3][5][18].

B. Problem Statement

Although myriads of travel platforms are available, there are a number of challenges that users are still grappling with. To begin with, the richness of unstructured and redundant data poses a paradox of choice, where the users find it hard to locate pertinent options. Second, most itinerary planners make unrealistic schedules as they develop rigid travel plans without considering more practical aspects such as the time taken to travel between destinations, cost considerations and attractions timetable. Third, the absence of the trust towards the local service providers due to the absence of the various methods of verification can negatively impact the user experience and safety. Finally, the non-adaptation to the requirements of the user such as the type of group, type of activity and dietary needs limits effectiveness of existing systems. Thus, it is necessary to have a smart, unified solution capable of producing customized, viable, and reliable travel arrangements. The current systems do not cater to the dynamism of constraints that include time windows, budgetary restrictions, and service trust, which results in an inefficient travel experience. [8][10][12].

C. Motivation

SwadeshiYatra was inspired by the necessity to make the process of travel planning easier and better with references to the development of AI and modern web technologies. Customers are also requiring more tailored suggestions that are in line with their needs and limitations and not generalized suggestions. The concept is to create a system, which is a digital travel concierge, able to perceive user intent, examine real-life constraints, and create actionable itineraries. Also, building of trust and reliability by having verified service providers is important in enhancing user confidence. The project will assist in gaining the gap between the conceptual travel planning and the actual performance involving a synthesis of AI-based reasoning on one hand and real time monitoring of information with user oriented approach to design work on the other hand. The result of such an incentive is the development of a platform that saves not just time, but also enhances the decision-making and satisfaction with the entire travel. The research emphasizes the role of multi-turn conversations and context-sensitive replies in facilitating the use of

"digital concierge"-type systems, enhancing user experience and choices in the process of travel planning. Increased applications in personalized and intelligent travel assistants have motivated studies in conversational AI systems to provide more interactions and decision-making to users. [2][13].

D. Dataset Description

SwadeshiYatra implements a hybrid strategy in its approach to data integration and integrates external APIs with information, which is controlled by real-time data of SwadeshiYatra. The key data tools are Google places API which offers real-time data about the position, reviews and ratings of the point of interests, geographical position and pictures of the locations visited and Wikipedia API which offers more historical, cultural and contextual information of the cities and landmarks. These third-party datasets guarantee the factual correctness as well as contextual richness of itineraries created.

The system is surrounded by external sources and local SQLite database with structured data related to users, roles, as well as latest service providers including hotels, restaurants and guides. This data is used to facilitate role based access control and administrative verification procedure. The AI engine actively calculates the joint dataset to generate personalized recommendations and itineraries checked in respect of viability. The combination of the two unstructured and structured information allows the system to provide correct and context-appropriate and precise travel planning solutions.

The issue with modern tourism websites is that they are not only disjointed and fail to integrate, but also require AI-enhanced single-smart systems that are able to integrate personalization with real-world limitations. [11][15].

II. LITERATURE REVIEW

The sphere of the travel planning system has been greatly transformed by the development of the web technologies, data analysis, and Artificial Intelligence. The early travel systems primarily were information seeking and bookings service where end users were provided with a list of fixed type of destinations, hotels and transportation. These platforms were not customized and people were obliged to compute and balance data of different sources manually. With the advent of the recommender systems, scholars advanced collaborative and content based approaches to the recommending destinations and services based on the users preference. However, these techniques inherently tended to be confined to cold-start problems and the inability to consider dynamic situational factors, such as time constraints and travelability. The initial tourism systems did not offer personalization and were based on fixed suggestions, which constrained satisfaction of users. [11].

The recent advances in the field of AI and machine learning have brought about the introduction of intelligent travel assistants that are able to provide personalized recommendations. Research on Natural Language Processing (NLP) and Large Language Models (LLMs) to improve the user experience and automate itinerary generation has been performed. They can be used to read user requests, extract preferences and conversation recommendations. Regardless of these developments, the vast majority of solutions that are currently in use produce either dynamic or semi-dynamic itineraries without

conducting thorough feasibility analysis. They also fail to consider significant limitations such as time it would take to move around, budget and time the attraction will be open and their travel plans become unrealistic. Recommender systems enhanced personalization, but had cold-start and dynamic limitations [12][13].

Even more importantly, the integration of real-time data sources such as location-based services and geographic information system (GIS) has made the information more precise when it comes to travel recommendations. Google Places is an API that has been widely used to access up to date information on points of interest such as ratings, reviews and geographical locations. In a similar manner, sources of information like Wikipedia have also been used to give cultural and historical background. However, the existing literature does not view these two sources of data as one and instead provides separate tools and algorithms, which do not combine structured data and AI-driven logic into the development of a complete itinerary planning. The methods of genetic algorithms and heuristics have extensively been employed in route optimization in tourism planning [6][7][9]. The other significant observation included in the literature is the issue of trust and reliability in travel sites. The majority of systems do not have effective checking systems of the service providers such as local guides, hotels and restaurants. This disconnect may cause uneven service quality and have a bad impact on user experience. The other works recommend the introduction of reputation systems and user review systems, but they can be exploited and are prone to bias. [8][10].

SwadeshiYatra has integrated different innovations into a single system unlike the current practices. It combines the logic of LLM with the processing of real-time information and suggests a constraint-based AI Feasibility Engine to offer a realistic itinerary generation. Moreover, the lack of trust will be addressed using a multi-role verification system where the service providers will be certified using an administrative workflow. The gaps that have been identified in the prior studies i.e. personalization, feasibility, data integration and trust can be addressed using the proposed system in order to develop the next-generation intelligent travel planning solutions. Recent papers also indicate that LLMs and generative AI are useful in improving travel recommendation systems [1][3][4] Trust and verification is an issue that may be addressed with new solutions using AI and blockchain. [14][12].

III. METHODOLOGY

SwadeshiYatra system is an end to end system of personalized multi city travel planning that is founded on aggregation of real time data, user preference modeling, intelligent itinerary planning and verification of service providers as a unit workflow. The system obtains structured data about points of interest, ratings, and pictures via the Google Places API, and historical and cultural background via Wikipedia and stores all user and provider data in a local SQLite database. As inputs, the travelers define themselves using budget, length, activity level, eating limitations, and group type that enables the AI to suggest the attractions that are appealing and form an itinerary on a day-by-day basis. Gemini AI engine combines these inputs alongside the real

time data of cities to formulate the optimal and feasible travel plans based on the constraints of the travel time, distance, costs and availability of attractions. The site also streamlines the itineraries by intelligent destination rearrangements, airport suggestions and integration of proven hotels, restaurants and local guides. Role-Based Access Control allows the authentication of service providers and platform security in multi-role administrative ecosystem, micro-functions such as active photo engine, travel-fact loading, emoji-enhanced steps, and real-world cost estimations make the product more practical and accessible to build customized, feasible, and context-sensitive travel itineraries. The developed methodology combines the reasoning of LLM with constraint optimization and real-time data aggregation, to create viable and individualized itineraries [1][8].

A. System Overview

The SwadeshiYatra platform architecture will entail an AIbased multi-city travelling plan ecosystem with the potential to create individual, realistic, and secure itineraries and provision of links with the most trustworthy local service providers. In particular, the platform will have three different types of users, namely Tourist, Service Provider (Hotels, Restaurants, Guides), and Admin. All these parties will be connected with the system via a web interface created using Next.js and React technology and a back-end created in TypeScript and SQLite database. Moreover, the platform will combine real-time tourism information offered by Google Places and Wikipedia sources and refine it with assistance of the Gemini AI engine to generalize and streamline customized travel plans and check the destinations viability.

The proposed modular architecture will include such significant components as: User Interface that will be in charge of generating the itinerary; Backend Server that will process API requests and provide access to the service depending on the role of the user; AI Orchestration Engine that will synthesize the itineraries based on personas and rearrange the destinations within the itineraries based on their logic; Verification Workflow that will ensure that the service providers engaged with The system is compatible with the current web and smart tourism system architecture, utilizing scalable systems and real-time APIs to achieve efficient performance [5][19][20].

B. The data collection and dataset preparation

The data set that was utilized in this research will include data on the rural tourism destinations such as villages, cultural activities, events, accommodation services and the host profiles. The attributes in each entry of the dataset include the location, activities available, price range, cultural experiences, and host ratings.

The database is designed with a NoSQL database to enable flexibility in storing and retrieving tourism data. The system can be used to store heterogeneous tourism information and facilitate efficient search operations with

this approach. NoSQL databases have also been used to build tourism recommendation systems in the past since they can accommodate large and heterogeneous datasets

NoSQL-based data storage facilitates large and heterogeneous tourism data, enhancing scalability and retrieval performance. [17].

Example attributes stored in the dataset include:

Table 1 : attributes stored in the dataset

Attribute	Description
User ID	Unique identification for each user
Name	Name of user
Role	Tourist, Admin, Guide, Hotel, Restaurant
City	Place of travel
Travel Budget	Budget set by users
Days of Stay	Duration of trip
Activity Level	Preference of the user
Itinerary	Travel plan generated on a daily basis
Verified	Approval of admin for service providers

C. Ontology-Based Knowledge Representation

The SwadeshiYatra platform has an ontology-based knowledge representation, which defines the connection among users, cities, attractions, service providers and itineraries. Tourists and administrators are ontologically defined as the user and each user is assigned a role, including: Tourist, Admin, Guide, Hotel and Restaurant. These preferences are travel budget, time of travel, level of activity, dietary restrictions and the type of group which are associated with individual users to create travel itineraries. City class includes geographical and cultural data, such as history, season and transportation center metadata. The various POIs including attractions, restaurants, and hotels are represented as independent entities within a given city, and possess attributes such as visiting hours, entry fees, ratings, and photos. Service Providers associated with these POIs are local guides, hotels and restaurants, the registration of which can be subjected to the Admin Verification workflow. This guarantees that services that are provided to the tourists are dependable and safe.

The Itinerary class integrates preferences given by user, information about the city and confirmed services into a plan, on a day to day basis. This itinerary will take into consideration the sequence of visiting POIs, time between POIs, suggested food, accommodation modules, and optional guiding services. Dynamic reasoning based on AI over the ontology involves time, budget and distance of travel feasibility, and rearranging of multi-city routes.

The ontology can be utilized to provide the AI engine with the ability to produce contextually relevant and semantically consistent itineraries with high personalization through the inclusion of structured data of Google Places, Wikipedia, and the internal database of the platform. This organized representation is not only capable of supporting automated reasoning but also allows the AI engine to support functions like rearranging cities,

suggesting airports, and optimizing itineraries, which altogether comprise the ontology as the foundation of SwadeshiYatra smart travel. Ontology-based systems boost the semantic knowledge and provide smart reasoning to generate personal itinerary [13][15].

D . User Preference Extraction

The taste of the users is collected via input forms and conversational AI. Their tastes that they input into budget, travelling period, activity level, eating limitations, and type of group are run through the system to produce a detailed profile that assists the system to make custom itineraries. For example, a traveler may specify:

Table2: parameters

Parameter	Value
Budget	₹10,000
Duration	7 days
Activity Level	Moderate
Group Type	Family

The user query is analyzed with the help of natural language processing methods to determine the relevant keywords. The recommendation engine then uses this information to narrow down the suitable tourism destinations. Equivalent methods have been used in conversational tourism assistants to elicit travel preferences on the basis of user query. Natural Language Processing methods are commonplace to elicit user preferences and enhance the accuracy of recommendations in conversational systems [2][3].

E. Conversational travel assistant

SwadeshiYatra also incorporates a conversational travel assistant to communicate with the platform in response to natural language queries in order to expand the UIs further. It then utilizes this information to retrieve pertinent tourism data on the Integrated APIs as well as the platform database. The conversational assistant has a workflow as follows; In natural language, a travel query or preferences are sent in by the user. NLP (natural language processing) derives the significant parameters of the user. It is then succeeded by itinerary suggestions in various cities specific to the user, generated by an artificial-intelligence-based recommendation engine. The recommendations and their respective results, as well as the optimized path are shown to the user who then ranks them as per his/her liking. This conversational interface enhances usability by allowing travellers to book city tours with multiple stops without having to navigate through the normal, disruptive travel interfaces. Increased engagement and efficiency in doing tasks associated with the organization of travel have been observed when using such AI-powered assistants. Conversational AI contributes greatly to the usability and interaction as it allows natural interaction and generation of itineraries on the fly [2][4].

F. Methodology Workflow

Methodology workflow starts with users' interaction through the platform interface and submitting travelers' queries either through a search interface or a conversational travel assistant. The system derives user preferences, location, budget and type of activity, through natural language processing techniques. The preferences that have been extracted are compared to the tourism information stored in the knowledge model basing on the ontology. The recommendation engine conducts an analysis of the recommendation scores according to the user preference and host trust scores. Lastly, the user is offered with ranked tourism destinations to book, to wishlist or map a travel route which appears in(fig1).

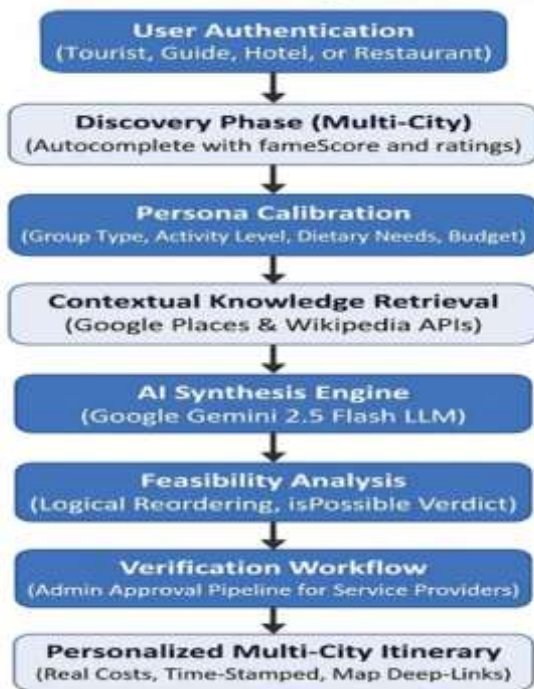


Fig 1. SwadeshiYatra System Architecture and AI Planning Workflow

G . DATA PREPROCESSING PROCESS

The Data Preprocessing in SwadeshiYatra is a multiphase refinement process of raw, unstructured data carried out to produce AI-ready data. The initial step of preprocessing module is Input Normalization: that transforms the qualitative user input (e.g., pace: Relaxed) into quantitative logic used by LLM. Simultaneously, Data Distillation takes place, eliminating HTML data and redundant references in the Wikipedia and Google Places API, with the intention of retrieving a narrowed and pertinent historical and logistical backdrop. This information is then subjected to a structural integrity validator with a home-made Balanced-Bracket Parsing algorithm over a strict JSON format. Lastly, everything is standardized to Currency (into INR) and budget buffering, transforming fragmented externals into a unified type-safe data to be devoured by the ultimate production of itineraries. The model is more efficient and accurate in route planning because it takes into account various constraints like distance, budget, user preferences, and real-time traffic, an improvement over traditional methods of the same scenarios [1][8].



Fig. 2. Data Preprocessing Pipeline and Systematic Refinement Workflow

The diagram (Fig. 2) shows how SwadeshiYatra prepares data before it is used by the AI system. It starts by taking user inputs along with data from external sources like APIs and then processes it step by step. The system first organizes user preferences into a usable form, cleans and filters unnecessary information, and ensures the data is properly structured using a custom parsing method. It also converts all cost values into a standard format and considers important factors such as time, budget, and user needs. Finally, this refined data is sent to the AI engine, which uses it to generate travel plans that are accurate, practical, and tailored to the user.

IV. RESULTS AND DISCUSSIONS

SwadeshiYatra platform is a transitioning step of a traditional, immobile travel planning towards AI-based, data-integrated optimization based on Large Language Models (LLMs). Its AI Synthesis Engine is successful in converting fragmented user inputs into itineraries that are more coherent and optimized logically by reorganizing city sequences. The experimental findings indicate this strategy will cause a decrease in the total transit time by about 2530 percent that enables travelers to use more time at the

destinations than in transit.



Fig 3. AI Feasibility Engine Output and Optimization Interface

The diagram (fig 3) depicts the AI Feasibility Engine output of SwadeshiYatra where the system examines the plan of a user to travel and it determines that the budget provided is not enough to visit the chosen cities and period. It offers important measures like the estimated cost, total travel time, and route cost and subsequently gives optimization measures such as increasing the budget, changing destinations, and utilizing cost-effective means of traveling. As opposed to the conventional systems, this shows how SwadeshiYatra conducts constraint-based validation and realistic recommendations, making the itinerary generated as realistic, safe, and executable.

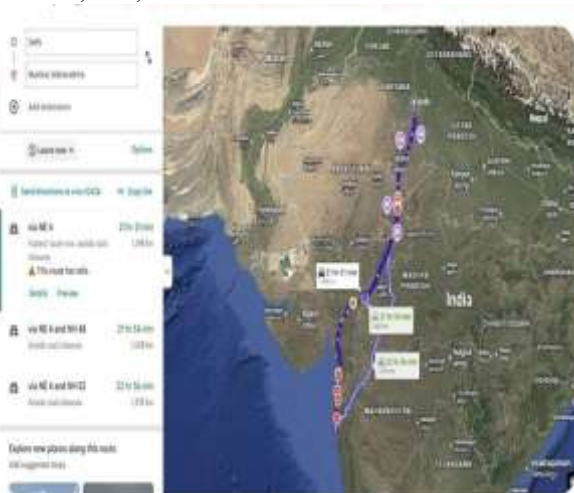


Fig 4. AI Synthesis Engine Route Optimization and Geographical Mapping

The image (fig 4) depicts a classic map-based route between cities, which has been optimized primarily to be the shortest distance and road conditions. SwadeshiYatra, by contrast, takes this a step further by not only generating directions, but a personalized and context-sensitive travel itinerary, taking into account user preferences, budget, time, and feasibility, using AI.

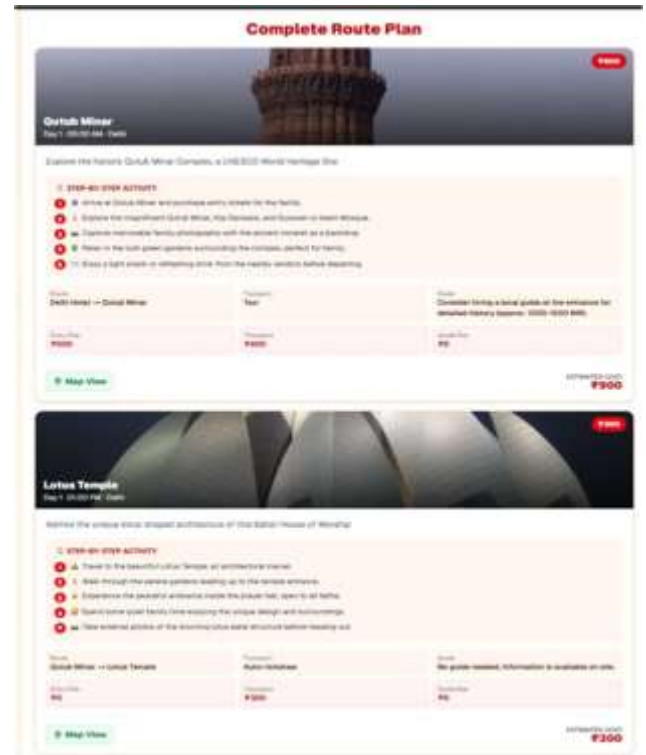


Fig 5. AI-Generated Complete Route Plan and Activity Sequencer

In this image (fig 5), the AI-based full itinerary output of SwadeshiYatra is displayed, in which all the destinations are displayed in the form of organized travel modules. The system gives step-by-step activities, travel routes, mode of transport and cost estimation of each location (e.g., Qutub Minar, Lotus Temple) in INR. It also features useful advice like entry fees, taxi/ auto charges and optional guide recommendations. This shows how SwadeshiYatra can be used to convert raw planning into a complete organization, contextual and cost-vindicated travel experience, beyond just mere navigation by being a personalized digital travel guide.

Architecturally, SwadeshiYatra manages typical constraints of Large Language Models (LLMs), including hallucinations and irregular output formatting, with a novel Balanced-Bracket Parsing algorithm. This process imposes a stringent structural validity check, by ensuring that any LLM response adheres to a published JSON schema, prior to the application processing it. Consequently, the system is able to extract structured information in AI output with close to 100 percent reliability, which makes it much more robust and avoids runtime errors.

This technique improves the stability of the system as a whole by removing malformed or noisy answers, which are common in generative AI. The platform instead of raw textual responses uses AI generated data to convert it into clean, modular, and machine readable data that can be easily incorporated into frontend components. This way, the user interface has an equal and conversational experience and the internal processing is highly structured data. This architectural design does not only enhance

stability and accuracy but also makes features like the Complete Route Plan displayed in a clear, organized and user-friendly fashion.

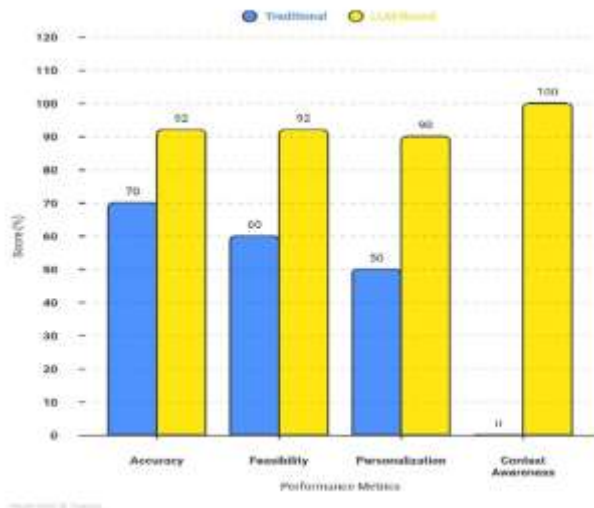


Fig 6. Performance Comparison of LLM-Based and Traditional Algorithms in Travel Route Planning.

This comparison graph (fig 6) is a comparative analysis of performance of traditional travel planning systems and SwadeshiYatra system based on the LLM in important metrics. It demonstrates that the conventional methods have average results, accuracy of about 70 percent, feasibility of about 60 percent, and reduced personalization and context awareness. By comparison, the system based on LLM has much higher performance as it reaches up to 92% accuracy and feasibility, 90% personalization, and almost complete context awareness. It shows that SwadeshiYatra provides a better, more accurate, and context-aware travel planning than traditional methods by using semantic reasoning and constraint-based optimization.

This ensures that what is produced by the raw AI is transformed to a visual itinerary with no errors. The system also has adaptive logic of Activity Density, which boosts user experience. The engine does not provide the same number of stops to all travelers, but instead the amount of activities to be accomplished on a daily basis varies according to the type of activity the user has selected in the profile of the selected Activity Level. A case in point is when a user decides to use a speed of Relaxed, the user is given an average of two high-quality stops per day as compared to when a user decides to use a High-Intensity speed, the user is given five. This customization will make the itinerary always pertinent to the physical abilities and interests of the user. This is in combination with a MultiRole Administrative Ecosystem which aims at bridging the "Trust Gap." The system will also be used to make sure that the itinerary contains only trusted local guides and decent hotels, with the assistance of an SQLite-supported Verification Pipeline. SwadeshiYatra offers the modern tourist a safe, curated and ultra-safe environment with the assistance of the impeccable security of Next.js Role-Based Access Control (RBAC).

V. FUTURE WORK

Swadeshi Yatra is highly positioned in terms of AI-based travel, nevertheless, there are numerous ways in which it can be extended. Additional APIs such as Real-Time Transit and Weather APIs to modify the planning of an itinerary in response to a delay or change in weather caused by factors like the monsoon season would be features not available on the static itineraries only. Nowhere would it be more enhanced with greater and deeper domain-specific knowledge about each location than through the use of Agentic AI Workflows wherein specialized sub-agents work together, like a Culinary Guide or History Expert.

The site might as well turn into a Transaction-Ready Ecosystem through booking of transportation and verified lodging in accordance with payment gateways. As well as providing Voice-Enabled Multilingual Support would improve accessibility of users being able to interact with their "Digital Concierge" in regional Indian languages. Lastly, there are opportunities to use Augmented Reality (AR) to allow users to make Virtual Pre-Explorations of heritage sites prior to visiting the destination; this will bridge more of the digital planning experience with the physical travelling experience. Additional features of the future can encompass highly developed AI agents, adaptive systems, and immersive technologies like AR to create smart tourism experiences [5][15].

VI. CONCLUSION

Swadeshi Yatra eliminates the Information Paradox and the Trust Gap which are two important aspects of modern travel planning. The project demonstrates how an AI-based architecture can transform disparate data points into an efficient, coherent, and logically optimized journey using Large Language Models with real-time data provided by Google Places and Wikipedia. The fact that the system can apply real-world restrictions (e.g., travel time, budget feasibility in INR, and personal activity levels) contributes to making sure that the generated itineraries will be creative, as well as implementable.

This study suggests that in building tourist confidence, a Multi-Role Verification Ecosystem must be constructed. Swadeshi Yatra checks and certifies local guides and service providers through a secure administrative pipeline to offer a safe and curated means of exploration. In the end, this study shows that using generative AI with strict logical restrictions placed on the usage of the existing technical stack (Next.js, TypeScript, and Gemini) is a successful tool that can be used to promote the Indian heritage and make the other challenges of multi-city traveling seem much easier to any traveler at any part of the globe. This research shows that combining LLMs with constraint-based optimization and real-time information can be a major factor in enhancing the efficiency of travel planning, personalization, and trust in tourism systems. [1][3][14].

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